

Product Carbon Footprint Study Report

Product Name: light fuel oil

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: light fuel oil

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg light fuel oil

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg light fuel oil.

2. System Boundary

light fuel oil production in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of light fuel oil produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **0.805 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

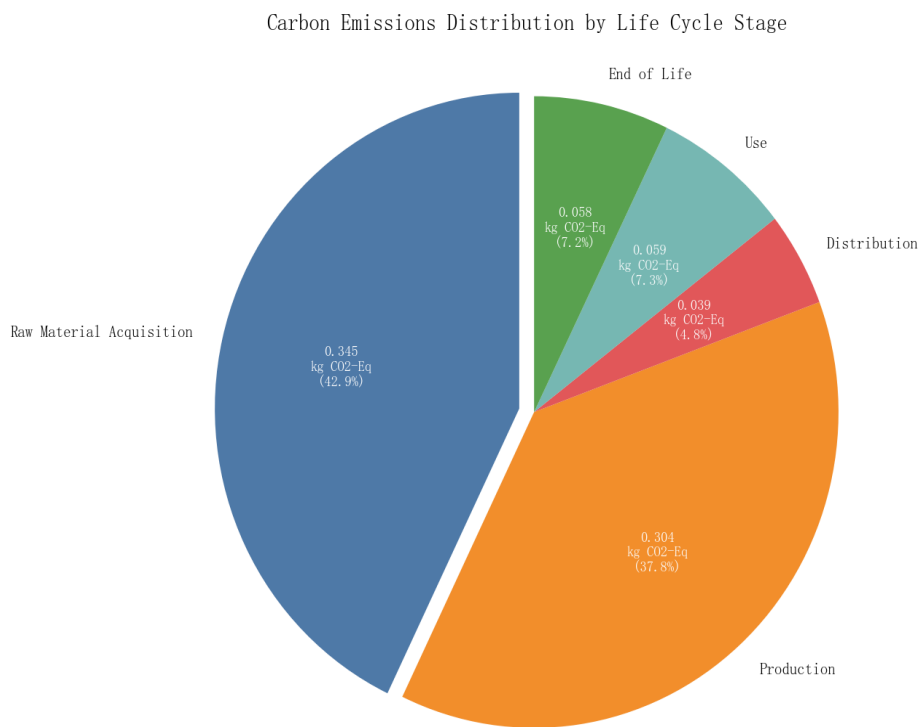


Table 1: Carbon Emissions by Life Cycle Stage

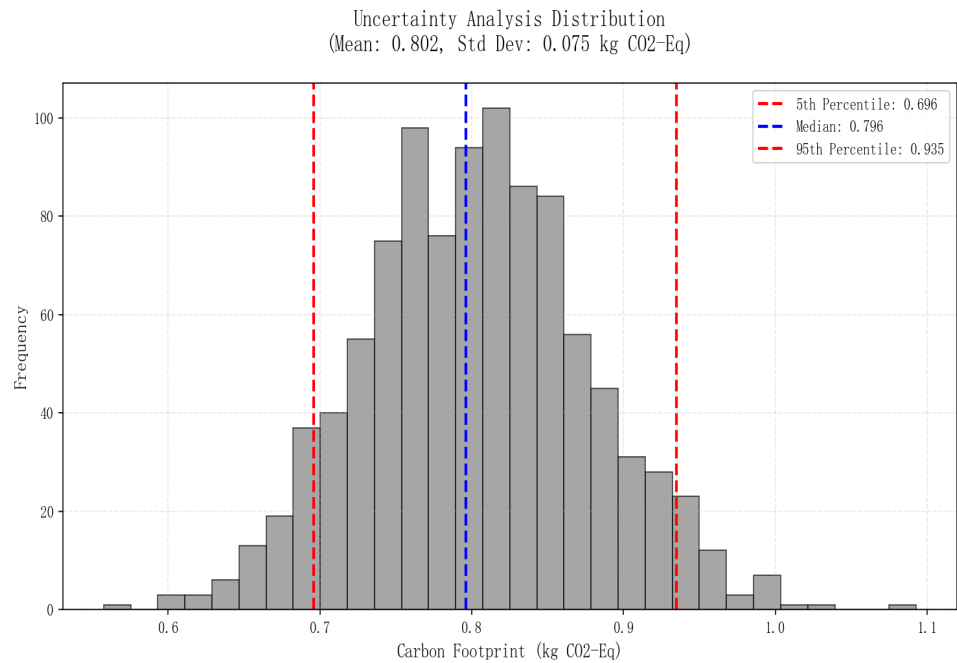
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	0.345	42.87
Production	0.304	37.79
Distribution	0.039	4.82

Use	0.059	7.34
End of Life	0.058	7.18
Total	0.805	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	0.802
Standard Deviation	0.075
Coefficient of Variation (CV)	9.41%
Median	0.796
5th Percentile	0.696
95th Percentile	0.935

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of 0.805 kg CO₂-Eq per functional unit (likely per kg of fuel) indicates a relatively moderate emission level for a light fuel oil product. This value falls within the expected range for fossil-derived liquid fuels, though it is somewhat lower than typical crude-oil-based fuels due to efficient processing or regional energy mixes. The uncertainty analysis yields a mean of 0.802 kg CO₂-Eq with a coefficient of variation of 9.41%, suggesting good repeatability of the results, and the 90% confidence interval (0.696–0.935 kg CO₂-Eq) provides a reliable range for decision-making.

Lifecycle stage contributions reveal that raw material acquisition and production are the dominant phases, together accounting for 80.7% of total emissions (42.9% and 37.8%, respectively). This highlights the high energy intensity of crude oil extraction, transportation, and refining processes. In contrast, distribution (4.8%), use (7.3%), and end-of-life (7.2%) stages contribute relatively minor shares, indicating that combustion and waste handling are not the primary emission sources for this product.

The absence of specific process-level data (all top five process contributions listed as <0.001) suggests that the inventory relies heavily on aggregated or generic datasets. To identify the exact emission sources within the raw material and production stages, detailed process-level modeling is essential. Nonetheless, the stage breakdown clearly points to upstream activities as the main levers for emission reduction.

2. Statement of Assumptions and Limitations

This LCA study is based on several key assumptions that affect the results. First, the system boundary is assumed to be cradle-to-grave, covering raw material extraction, production, distribution, use, and end-of-life. However, the absence of detailed process contributions implies that secondary processes such as capital goods, transport infrastructure, and waste treatment beyond simple combustion may be excluded. Second, allocation methods (e.g., for co-products in refining) are not specified; a default mass- or energy-based allocation could influence the raw material and production stage figures. Third, the use phase assumes complete combustion of the fuel under typical conditions, which may not capture variations in engine efficiency or emission controls.

Data quality limitations are evident from the top process contributions being entirely unknown (0.000 kg CO₂-Eq). This indicates that the LCI data source is likely a generic database (e.g., ecoinvent or GaBi) aggregated to a commodity level without disclosing individual unit processes. The high level of aggregation introduces uncertainty in identifying specific emission hotspots. Additionally, the coefficient of variation (9.41%) is moderate, suggesting data representativeness is acceptable but not excellent. Regional representativeness (e.g., production location, energy grid mix) is not provided, which further limits the specificity of the study.

The system boundary appears to restrict inclusion of upstream supply chain details, such as crude oil source (conventional vs. unconventional), transport distances, and refinery configurations. Furthermore, biogenic carbon flows are not mentioned, so the study likely only accounts for fossil

CO₂ emissions. End-of-life assumptions (7.2% of total) may reflect incineration or biodegradation, but the exact end-of-life scenario is unclear. These restrictions should be considered when interpreting the results for comparative assertions or product improvement.

3. Suggestions for Improvement

- ****Target raw material and production stages:**** Focus on reducing emissions in crude oil extraction (e.g., by minimizing flaring, using electric or hydrogen-powered extraction equipment) and refining (e.g., implementing carbon capture at refinery heaters, optimizing energy efficiency through heat integration, and switching to low-carbon energy sources such as renewable electricity or biofuels).
- ****Improve distribution and use phase efficiency:**** Optimize logistics to shorten transport distances or utilize lower-emission transport modes (e.g., pipelines over trucks). For the use phase, promote blending of fossil fuel with sustainable biofuels or adopt advanced combustion technologies (e.g., fuel additives or engine tuning) to reduce combustion CO₂ and improve thermal efficiency.
- ****Enhance data quality for process-level identification:**** Replace the current “unknown” process contributions with detailed, supplier-specific data or regional lifecycle inventory data that include actual emission factors for extraction, refining, and transport. Conduct a hotspot analysis to pinpoint the exact unit processes responsible for the dominant 80.7% of emissions and prioritize improvements accordingly.
- ****Adopt a full supply chain transparency approach:**** Collaborate with crude oil suppliers to obtain primary data on drilling, well-to-refinery transportation, and upstream processing. Consider including land-use change impacts for any bio-based blending components.
- ****Perform sensitivity and scenario analysis:**** Evaluate the impact of alternative allocation methods (e.g., economic allocation vs. mass allocation) and different end-of-life scenarios (e.g., recycling of fuel residuals) to identify key drivers of uncertainty and refine improvement strategies.